

Erdős Problem #36

EP-36

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Find the optimal constant $c > 0$ such that the following holds.

For all sufficiently large N , if $A \sqcup B = \{1, \dots, 2N\}$ is a partition into two equal parts, so that $|A| = |B| = N$, then there is some x such that the number of solutions to $a - b = x$ with $a \in A$ and $b \in B$ is at least cN .

Source and attribution. Statement reproduced from UnsolvedMath, whose official dataset is released under the CC BY 4.0 license.

Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>